

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Water/H ₂ O	1			1		
	(b)	(i)		It is the active site (1) It is complementary in shape to the cytochrome c (1) OR {Genetic code/DNA base sequence} for polypeptide chains similar in all organism/Has not mutated (1) So amino acid sequence/polypeptide chains similar/1 ^o structure similar (1)	2			2		
		(ii)		It changes shape (1) So no longer complimentary shape to the active site/does not fit into active site any more/can no longer form enzyme substrate complexes (1)		2		2		
	(c)			{Increases/becomes} {more alkaline/less acidic} (1) Less {H ⁺ /protons} present as they are not {pumped/transported} (into the intermembrane space) (1)		2		2		
	(d)	(i)		Protons do not bind to electrons and oxygen (1) Electron carrier system stops/reduced NAD can't pass on the protons (1) MUST BE IN CONTEXT OF WHEN INHIBITOR IS ADDED			2	2		
		(ii)		Any 2× (1) from: (Lactate is end product) of anaerobic respiration(1) pyruvate is reduced to regenerate NAD/pyruvate is reduced by using reduced NAD (1) for glycolysis to continue (1)			2	2		
				Question 6 total	3	4	4	11	0	0